

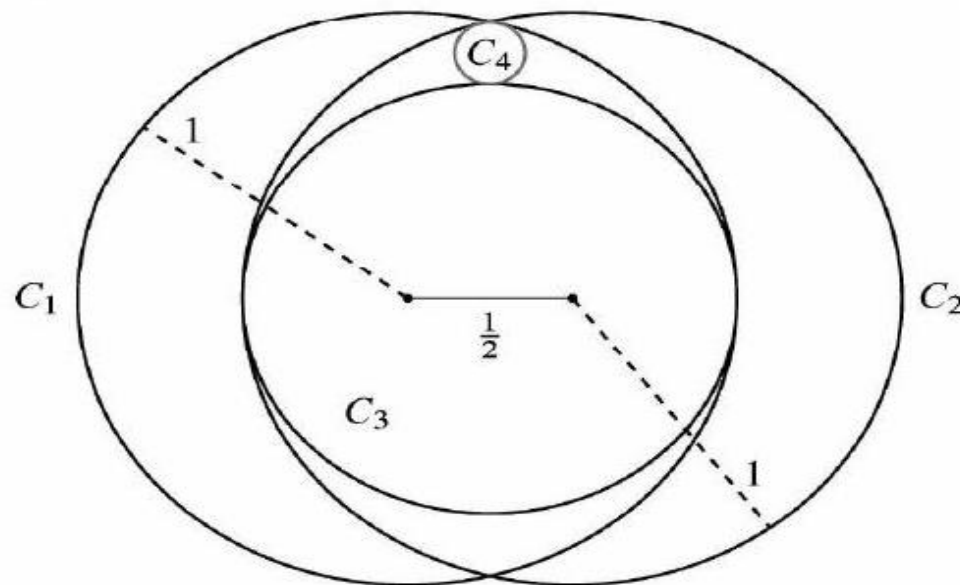
2023 AMC 10A Problem 22 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10A Problem 22. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10A Problem 22

Problem:

Circles C_1 and C_2 each have radius 1, and the distance between their centers is $\frac{1}{2}$. Circle C_3 is the largest circle internally tangent to both C_1 and C_2 . Circle C_4 is internally tangent to both C_1 and C_2 and externally tangent to C_3 . What is the radius of C_4 ?



Answer Choices:

A. $\frac{1}{14}$

B. $\frac{1}{12}$

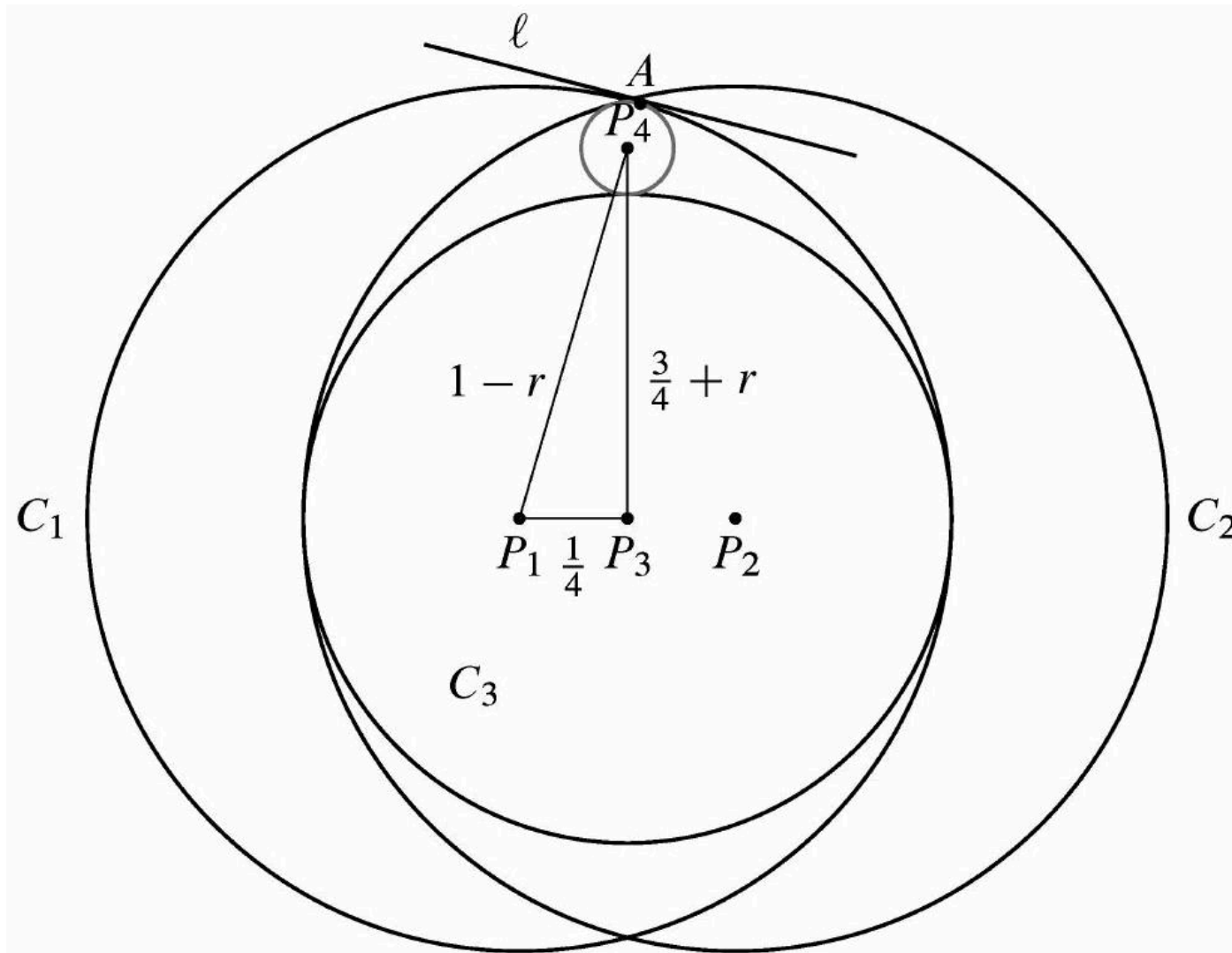
C. $\frac{1}{10}$

D. $\frac{3}{28}$

E. $\frac{1}{9}$

Solution:

Let r be the radius of C_4 . Let the point P_k be the center of C_k for $1 \leq k \leq 4$. Let A be the point of intersection of circles C_1 and C_4 and let ℓ be the tangent line to P_1 at A . By symmetry, P_3 is the midpoint of $\overline{P_1 P_2}$, and $P_1 P_3 = \frac{1}{4}$. Also by symmetry, P_4 lies on the perpendicular bisector of $\overline{P_1 P_2}$. The radius $\overline{P_1 A}$ is perpendicular to ℓ . Likewise the radius $\overline{P_4 A}$ is perpendicular to ℓ , so P_1 , P_4 , and A all lie on the segment $\overline{P_1 A}$ with P_4 between P_1 and A . Then $P_1 P_4 = 1 - r$.



The radius of C_3 is $\frac{3}{4}$. Then $P_3P_4 = \frac{3}{4} + r$, and $\triangle P_1P_3P_4$ is a right triangle with $P_1P_3 = \frac{1}{4}$, $P_3P_4 = \frac{3}{4} + r$, and $P_1P_4 = 1 - r$. By the Pythagorean Theorem,

$$\left(\frac{1}{4}\right)^2 + \left(\frac{3}{4} + r\right)^2 = (1 - r)^2$$

Expanding gives

$$\frac{1}{16} + \frac{9}{16} + \frac{3r}{2} + r^2 = 1 - 2r + r^2$$

which gives $r = \text{(D)} \boxed{\frac{3}{28}}$.

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](#) 

Powered by [Wiki.js](#)